# Block practice · Reproduction and development

Work through each question type in order. Each section has a worked-example retry and all further practice. Attempt only the part outlined in red.

State the role of a named reproductive hormone

Identify reproductive organs from their roles or positions

Compare haploid and diploid cells and explain why meiosis matters

Name contraceptive methods, explain how they work and compare them

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# Practice workbook · State the role of a named reproductive hormone

Functions of FSH, LH, oestrogen, progesterone and testosterone in the cycle or development.

6 question parts: 1 worked-example retry and 5 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-STATE-REPRODUCTIVE-HORMONE-ROLE

# 1. Question 1 · part b.ii · 2 marks

June 2025 Paper 2

Worked-example retry · Attempt part b.ii only.

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**1** (a) The words in the box give the names of some of the parts of the male and female reproductive systems.

| | | | |
|---|---|---|---|
| epididymis | ovary | oviduct | seminal vesicle |
| testes | uterus | urethra | vagina |
| vulva | | | |

Complete the table by giving the name of the part that matches the description.

**(7)**

| Description | Name of part |
|---|---|
| duct that carries urine and sperm | |
| produces sperm | |
| where fertilised ovum is implanted | |
| produces the fluid part of semen | |
| where sperm is deposited during intercourse | |
| where fertilisation happens | |
| releases ova | |

(b) Oxytocin and LH (luteinising hormone) are two hormones found in females.

(i) Describe the functions of oxytocin.

**(3)**

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(ii) Describe the functions of LH.

**(2)**

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**(Total for Question 1 = 12 marks)**

## 1. Question 1 · part b.ii · 2 marks · Working and answer

Working and answer

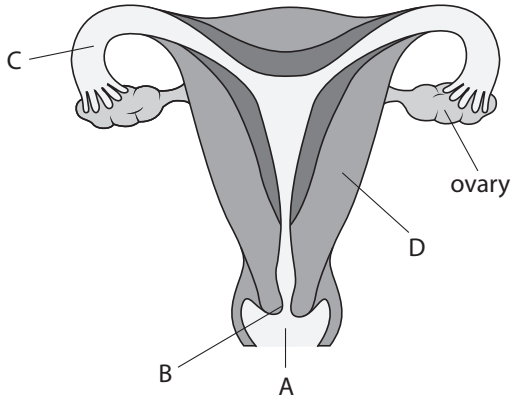
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# 2. Question 2 · part a.iii · 3 marks

January 2020 Paper 1

Further practice · Attempt part a.iii only.

**2** (a) The diagram shows some structures of the female reproductive system.




(i) Where are sperm deposited during sexual intercourse?

**(1)**

- ☒ **A**
- ☒ **B**
- ☒ **C**
- ☒ **D**

(ii) Which part of the female reproductive system contracts during birth to push the fetus downwards?

**(1)**

- ☒ **A**
- ☒ **B**
- ☒ **C**
- ☒ **D**

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# 2. Question 2 · part a.iii · 3 marks

January 2020 Paper 1

Further practice · Attempt part a.iii only.

(iii) The ovaries produce female sex hormones.

The box lists words associated with the female reproductive system.

| | | | | |
|---|---|---|---|---|
| FSH | wall | fertilisation | oestrogen | ovulation |
| lining | ADH | follicles | menstruation | |

Use words from the box to complete the passage about the ovaries.

**(3)**

The ovaries produce a hormone called ..... .

This hormone repairs the uterus ..... following menstruation. Once a month, one egg is released from the ovaries. This is known as ..... .

(b) State why a pregnant female should have more calcium in her diet than a female who is not pregnant.

**(1)**

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.....

(c) Some women breastfeed their babies.

Give one advantage of breastfeeding babies rather than feeding them bottled milk.

**(1)**

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## 2. Question 2 · part a.iii · 3 marks · Working and answer

Working and answer

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## 3. Question 8 · part a.i · 1 mark

January 2023 Paper 2

Further practice · Attempt part a.i only.

**8** (a) When women cannot conceive naturally, they can have fertility treatment.

Women having fertility treatment can be given the hormones FSH and LH.

(i) Give a role of LH in women.

**(1)**

(ii) Giving birth to twins or triplets is described as a multiple birth.

Explain how the use of FSH to treat infertility could increase the chances of a multiple birth.

**(3)**

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## 3. Question 8 · part a.i · 1 mark · Working and answer

Working and answer

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# 4. Question 6 · part f · 1 mark

June 2023 Paper 2R

Further practice · Attempt part f only.

**6** Read the passage below.

Use the information in the passage and your own knowledge to answer the questions that follow.

Pregnancy begins with the formation of a zygote. This stage is known as conception. The zygote develops into an embryo. By day five the embryo is transformed into a blastocyst, a hollow ball of about 100 genetically identical cells. These cells have not yet differentiated and they can provide a source of embryonic stem cells for use in medical research. Embryos produced following IVF treatment can be used as a source of embryonic stem cells.

In the second week following conception, the blastocyst begins the process of implantation and releases a hormone, called hCG. This hormone maintains the corpus luteum, a structure formed in the ovary following ovulation. After implantation the amniotic sac forms, which is where the blastocyst and later the fetus develop.

In the fourth and fifth week after conception, the umbilical cord forms. This connects the embryo to the placenta by week eight. The placenta is not fully developed until week ten when it begins to produce progesterone and oestrogen. All organ systems are present at this time in pregnancy.

Fetal development begins at about month four and it is possible to distinguish males from females at this time. In males the testes, which develop in the abdominal cavity, release testosterone. Female characteristics are produced in the fetus if testosterone is not released at the stage.

Hormones are important in starting labour and the birth process, about nine months after conception. The cervix dilates, with the head of the baby pushing against the cervix to help the process of dilation. Once the child is born and breathing normally, the umbilical cord is tied and cut, separating the child from the placenta.

(a) Describe the process that occurs at conception. (lines 1 and 2)

**(2)**

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# 4. Question 6 · part f · 1 mark

June 2023 Paper 2R

Further practice · Attempt part f only.

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(b) Describe the process that forms a blastocyst. (lines 2 to 4)

**(4)**

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(c) Which is a reason why cells taken from a blastocyst can be useful in scientific research?

**(1)**

- ☒ **A** the cells are already differentiated
- ☒ **B** the cells are able to differentiate into both sex cells and body cells
- ☒ **C** the cells are able to differentiate into any type of sex cell
- ☒ **D** the cells are able to differentiate into any type of body cell

## 4. Question 6 · part f · 1 mark

June 2023 Paper 2R

Further practice · Attempt part f only.

(d) Explain why it is important that hCG maintains the corpus luteum. (lines 7 to 9)

**(2)**

(e) Explain the importance of the development that takes place in the fourth and fifth week after conception. (lines 12 and 13)

**(2)**

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(f) Suggest the role of testosterone in the development of the fetus. (lines 17 to 19)

**(1)**

(g) Which change occurs during the birth process? (line 20)

**(1)**

- ☒ **A** oestrogen levels decrease
- ☒ **B** progesterone levels increase
- ☒ **C** release of oxytocin from the pituitary gland
- ☒ **D** release of hCG from the placenta

**(Total for Question 6 = 13 marks)**

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## 4. Question 6 · part f · 1 mark · Working and answer

Working and answer

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## 5. Question 6 · part a · 3 marks

June 2024 Paper 1R

Further practice · Attempt part a only.

**6** (a) The table gives the names and functions of hormones in the menstrual cycle.

Complete the table by placing one tick (✓) in each row of the table to show the main function of each hormone in the menstrual cycle.

**(3)**

| Hormone | Function of hormone | | | |
|---|---|---|---|---|
| | causes ovulation | builds up uterus lining after menstruation | maintains uterus lining during pregnancy | stimulates a follicle to mature |
| progesterone | | | | |
| FSH | | | | |
| oestrogen | | | | |

(b) The chance of pregnancy can be decreased by using contraception.

State **one** method of barrier contraception.

**(1)**

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(c) In-vitro fertilisation (IVF) can be used to increase the chances of pregnancy.

Describe the process of IVF.

**(4)**

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## 5. Question 6 · part a · 3 marks · Working and answer

Working and answer

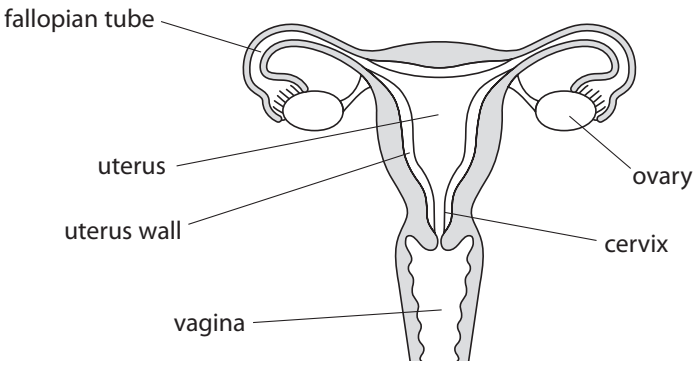
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# 6. Question 2 · part a.iv · 1 mark

June 2024 Paper 2R

Further practice · Attempt part a.iv only.

**2** (a) The diagram shows the female reproductive system.




(i) Complete the table by placing one tick (✓) in each row to show the role of each part of the female reproductive system.

**(3)**

| Part | Role | | |
|---|---|---|---|
| | where sperm are deposited | where eggs are produced | where fertilisation takes place |
| fallopian tube | | | |
| ovary | | | |
| vagina | | | |

(ii) Describe the action of the muscles in the uterus wall during labour.

**(2)**

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# 6. Question 2 · part a.iv · 1 mark

June 2024 Paper 2R

Further practice · Attempt part a.iv only.

(iii) Fertilisation takes place in the female reproductive system.

Which of these describes fertilisation?

**(1)**

- ☒ **A** a sperm cell fuses with an egg cell to form a haploid cell
- ☒ **B** a sperm cell fuses with an egg cell to form an embryo
- ☒ **C** a sperm cell fuses with an egg cell to form a zygote
- ☒ **D** a sperm cell fuses with an egg cell to form a baby

(iv) Before fertilisation can take place, ovulation needs to occur.

Name the hormone that causes ovulation.

**(1)**

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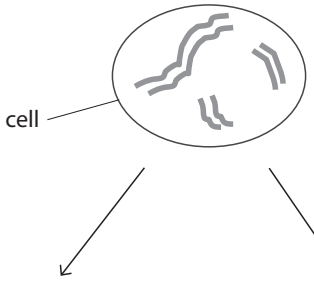
# 6. Question 2 · part a.iv · 1 mark

June 2024 Paper 2R

Further practice · Attempt part a.iv only.

(b) Following fertilisation and during pregnancy, cell division by mitosis takes place.

The diagram shows a cell with only three pairs of chromosomes instead of a full set.




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(i) Complete the diagram to show the cells formed when this cell divides by mitosis.

**(2)**

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P 7 4 4 9 5 A 0 8 2 8

## 6. Question 2 · part a.iv · 1 mark · Working and answer

Working and answer

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# Practice workbook · Identify reproductive organs from their roles or positions

Male and female reproductive system organs located on a diagram or matched to a stated job.

8 question parts: 1 worked-example retry and 7 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-MATCH-REPRODUCTIVE-ORGAN-TO-ROLE

# 1. Question 1 · part a · 7 marks

June 2025 Paper 2

Worked-example retry · Attempt part a only.

**1** (a) The words in the box give the names of some of the parts of the male and female reproductive systems.

| | | | |
|---|---|---|---|
| epididymis | ovary | oviduct | seminal vesicle |
| testes | uterus | urethra | vagina |
| vulva | | | |

Complete the table by giving the name of the part that matches the description.

**(7)**

| Description | Name of part |
|---|---|
| duct that carries urine and sperm | |
| produces sperm | |
| where fertilised ovum is implanted | |
| produces the fluid part of semen | |
| where sperm is deposited during intercourse | |
| where fertilisation happens | |
| releases ova | |

(b) Oxytocin and LH (luteinising hormone) are two hormones found in females.

(i) Describe the functions of oxytocin.

**(3)**

(ii) Describe the functions of LH.

**(2)**

**(Total for Question 1 = 12 marks)**

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## 1. Question 1 · part a · 7 marks · Working and answer

Working and answer

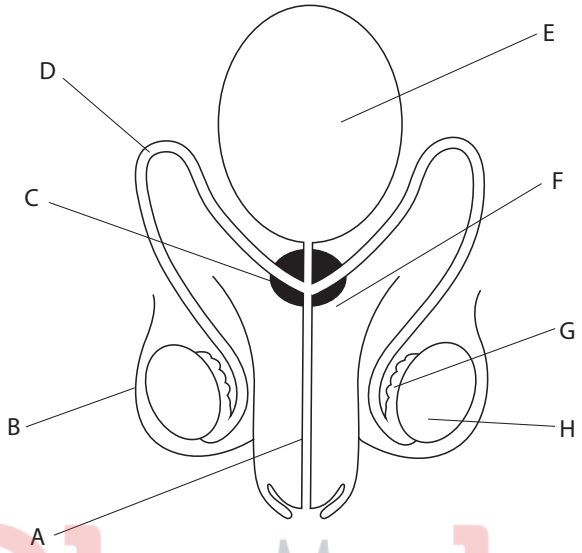
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## 2. Question 7 · part a · 4 marks

June 2019 Paper 2R

Further practice · Attempt part a only.

**7** (a) The diagram shows the male reproductive system.




The table gives descriptions of structures in the male reproductive system.

Complete the table by giving a letter from the diagram to match each description.

Each letter may be used once, more than once or not at all.

**(4)**

| Description of the structure | Letter |
|---|---|
| stores sperm | |
| produces sperm | |
| produces fluid to help sperm move | |
| contains cells dividing by meiosis | |

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P 6 0 1 8 7 A 0 1 8 2 4

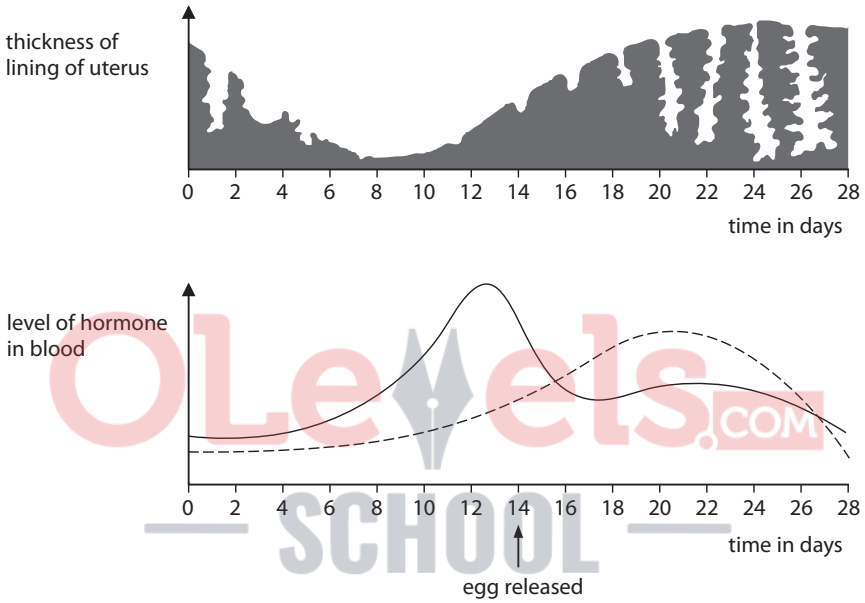
## 2. Question 7 · part a · 4 marks

June 2019 Paper 2R

Further practice · Attempt part a only.

(b) Changes occur in the female body during the menstrual cycle.

The graphs show changes in the lining of the uterus and the levels of two hormones that control these changes.




(i) Use information from the graphs to explain the cause of growth in the lining of the uterus after day 12.

**(2)**

(ii) Explain why fertilisation is less likely to occur from intercourse between days 8 and 12.

**(2)**

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## 2. Question 7 · part a · 4 marks · Working and answer

Working and answer

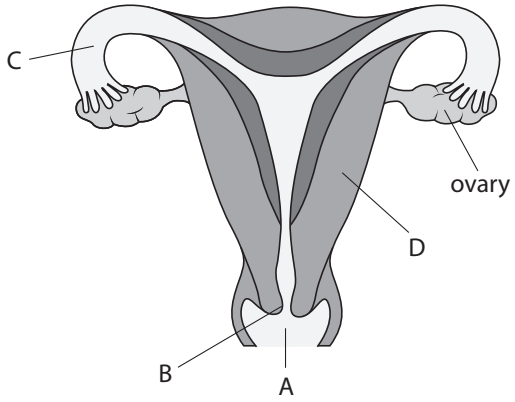
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# 3. Question 2 · part a.i · 1 mark

January 2020 Paper 1

Further practice · Attempt part a.i only.

**2** (a) The diagram shows some structures of the female reproductive system.




(i) Where are sperm deposited during sexual intercourse?

**(1)**

- ☒ **A**
- ☒ **B**
- ☒ **C**
- ☒ **D**

(ii) Which part of the female reproductive system contracts during birth to push the fetus downwards?

**(1)**

- ☒ **A**
- ☒ **B**
- ☒ **C**
- ☒ **D**

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## 3. Question 2 · part a.i · 1 mark · Working and answer

Working and answer

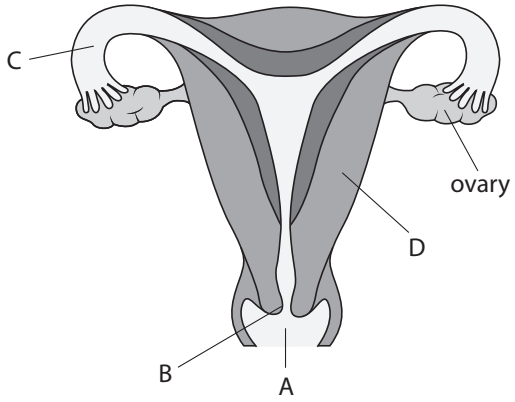
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# 4. Question 2 · part a.ii · 1 mark

January 2020 Paper 1

Further practice · Attempt part a.ii only.

**2** (a) The diagram shows some structures of the female reproductive system.




(i) Where are sperm deposited during sexual intercourse?

**(1)**

- ☒ **A**
- ☒ **B**
- ☒ **C**
- ☒ **D**

(ii) Which part of the female reproductive system contracts during birth to push the fetus downwards?

**(1)**

- ☒ **A**
- ☒ **B**
- ☒ **C**
- ☒ **D**

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# 4. Question 2 · part a.ii · 1 mark

January 2020 Paper 1

Further practice · Attempt part a.ii only.

(iii) The ovaries produce female sex hormones.

The box lists words associated with the female reproductive system.

| | | | | |
|---|---|---|---|---|
| FSH | wall | fertilisation | oestrogen | ovulation |
| lining | ADH | follicles | menstruation | |

Use words from the box to complete the passage about the ovaries.

**(3)**

The ovaries produce a hormone called ..... .

This hormone repairs the uterus ..... following menstruation. Once a month, one egg is released from the ovaries. This is known as ..... .

(b) State why a pregnant female should have more calcium in her diet than a female who is not pregnant.

**(1)**

.....

.....

(c) Some women breastfeed their babies.

Give one advantage of breastfeeding babies rather than feeding them bottled milk.

**(1)**

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## 4. Question 2 · part a.ii · 1 mark · Working and answer

Working and answer

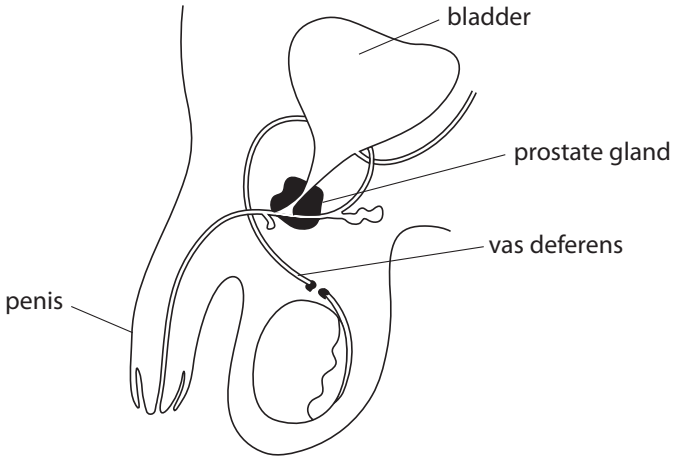
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# 5. Question 6 · part b.i · 1 mark

January 2022 Paper 1

Further practice · Attempt part b.i only.

**6** The diagram shows the reproductive organs of a male who has had a vasectomy.




(a) (i) Explain why having a vasectomy is an effective method of contraception.

**(4)**

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## 5. Question 6 · part b.i · 1 mark

January 2022 Paper 1

Further practice · Attempt part b.i only.

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(ii) Explain why this method of contraception does not prevent the spread of HIV which can cause AIDS.

**(3)**

(b) (i) Add an X to the diagram to show where haploid cells are produced.

**(1)**

(ii) Explain the importance of haploid cells in the process of fertilisation.

**(4)**

**(Total for Question 6 = 12 marks)**

## 5. Question 6 · part b.i · 1 mark · Working and answer

Working and answer

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## 6. Question 1 · part whole · 6 marks

June 2023 Paper 2

Further practice · Attempt part whole only.

**1** The box lists structures that may be found in the human body.

| | | | | |
|---|---|---|---|---|
| epididymis | oviduct | prostate | testis | |
| uterus | ureter | urethra | vagina | vulva |

Complete the table using words from the box to give the missing information.

**(6)**

| Description | Structure |
|---|---|
| produces sperm | |
| site of fertilisation | |
| site where sperm is deposited during intercourse | |
| place where zygote implants | |
| tube that carries sperm and urine | |
| produces the fluid part of semen | |

**(Total for Question 1 = 6 marks)**

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## 6. Question 1 · part whole · 6 marks · Working and answer

Working and answer

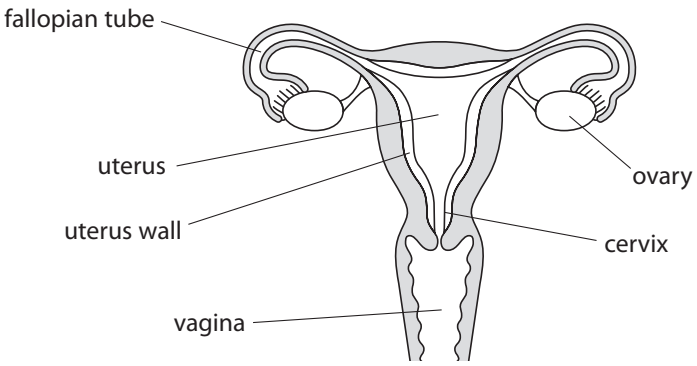
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# 7. Question 2 · part a.i · 3 marks

June 2024 Paper 2R

Further practice · Attempt part a.i only.

**2** (a) The diagram shows the female reproductive system.




(i) Complete the table by placing one tick (✓) in each row to show the role of each part of the female reproductive system.

**(3)**

| Part | Role | | |
|---|---|---|---|
| | where sperm are deposited | where eggs are produced | where fertilisation takes place |
| fallopian tube | | | |
| ovary | | | |
| vagina | | | |

(ii) Describe the action of the muscles in the uterus wall during labour.

**(2)**

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## 7. Question 2 · part a.i · 3 marks · Working and answer

Working and answer

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## 8. Question 1 · part a · 6 marks

November 2024 Paper 2

Further practice · Attempt part a only.

**1** (a) The information in the box gives structures or hormones involved in reproduction.

| | | | | |
|---|---|---|---|---|
| cervix | FSH | ovary | oviduct | ovum |
| oxytocin | progesterone | uterus | vagina | vulva |

The table lists some of the functions of these structures and hormones.

Complete the table by giving the missing information.

**(6)**

| Function | Name of structure or hormone |
|---|---|
| transfers ova to uterus | |
| contracts during birth | |
| releases ova | |
| dilates to allow birth | |
| maintains uterus lining during pregnancy | |
| causes contractions during birth | |

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## 8. Question 1 · part a · 6 marks

November 2024 Paper 2

Further practice · Attempt part a only.

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(b) There are various methods of birth control.

(i) Complete the table by giving an example of each method.

**(3)**

| Method of birth control | Example |
|---|---|
| barrier | |
| natural | |
| hormonal | |

(ii) Name one other method of birth control not given in the table.

**(1)**

.....

**(Total for Question 1 = 10 marks)**

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## 8. Question 1 · part a · 6 marks · Working and answer

Working and answer

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Practice workbook · Compare haploid and diploid cells and explain why meiosis matters

Chromosome numbers, homologous pairs and DNA mass in body cells and gametes, karyotypes, differences from mitosis, and why halving and variation in meiosis are important.

11 question parts: 1 worked-example retry and 10 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-COUNT-HAPLOID-DIPLOID-CHROMOSOMES

# 1. Question 2 · part c · 2 marks

June 2025 Paper 2R

Worked-example retry · Attempt part c only.

**2** (a) Cells are made up of different components. Each component has a particular function.

Identify the cell component that transports substances inside the cell.

**(1)**

- ☒ **A** cell membrane
- ☒ **B** cytoplasm
- ☒ **C** endoplasmic reticulum
- ☒ **D** ribosomes

(b) Mitosis is a type of cell division.

A student uses a light microscope to view a cell undergoing four stages in mitosis.

The diagrams, W, X, Y and Z, show the student's observations.



W X Y Z

(Source: https://www.pinterest.co.uk/pin/812547957739062607/)

(i) Give the letters to show the correct order of the diagrams in the process of mitosis.

**(1)**

(ii) State one function of mitosis in body cells.

**(1)**

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# 1. Question 2 · part c · 2 marks

June 2025 Paper 2R

Worked-example retry · Attempt part c only.

DO NOT WRITE IN THIS AREA

(c) Meiosis is another type of cell division.

Give two differences between the cells produced by meiosis and the cells produced by mitosis.

**(2)**

1 .....

.....

2 .....

.....

**(Total for Question 2 = 5 marks)**

Working and answer

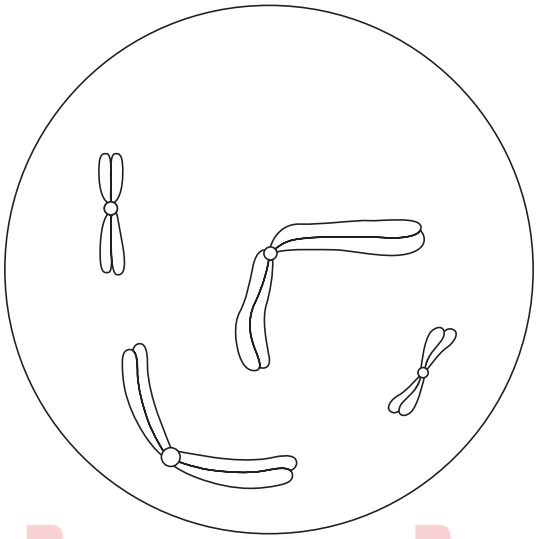
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## 2. Question 5 · part a.ii · 1 mark

June 2021 Paper 2

Further practice · Attempt part a.ii only.

**5** The diagram shows a cell in the early stages of mitosis.



(a) (i) Which molecule is found in the chromosomes? **(1)**

- ☒ **A** DNA
- ☒ **B** glucose
- ☒ **C** glycogen
- ☒ **D** mRNA

(ii) Draw a circle around two homologous chromosomes on the diagram. **(1)**

(iii) State how the diagram shows that this is not a human cell. **(1)**

.....

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## 2. Question 5 · part a.ii · 1 mark · Working and answer

Working and answer

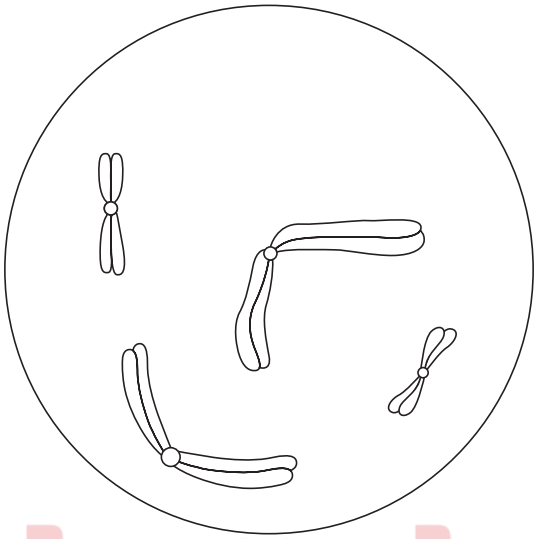
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## 3. Question 5 · part a.iii · 1 mark

June 2021 Paper 2

Further practice · Attempt part a.iii only.

**5** The diagram shows a cell in the early stages of mitosis.



(a) (i) Which molecule is found in the chromosomes? **(1)**

- ☒ **A** DNA
- ☒ **B** glucose
- ☒ **C** glycogen
- ☒ **D** mRNA

(ii) Draw a circle around two homologous chromosomes on the diagram. **(1)**

(iii) State how the diagram shows that this is not a human cell. **(1)**

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## 3. Question 5 · part a.iii · 1 mark

June 2021 Paper 2

Further practice · Attempt part a.iii only.

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(b) Describe what happens to the chromosomes during the four stages of mitosis.

**(5)**

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**(Total for Question 5 = 8 marks)**

## 3. Question 5 · part a.iii · 1 mark · Working and answer

Working and answer

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# 4. Question 5 · part c · 4 marks

January 2022 Paper 1

Further practice · Attempt part c only.

**5** (a) Describe the possible effects of a DNA mutation.

**(3)**

(b) Changes to whole chromosomes can happen when meiosis does not occur properly.

These changes are called chromosome aberrations.

The table shows the probability of children being born with a chromosome aberration to mothers of different ages.

| **Age of mother in years** | 15–19 | 20–24 | 25–29 | 30–34 | 35–39 | 40–44 |
|---|---|---|---|---|---|---|
| **Probability of chromosome aberration** | 0.0010 | 0.0010 | 0.0010 | 0.0015 | 0.0045 | 0.0140 |

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P 6 7 0 5 9 A 0 1 4 2 4

# 4. Question 5 · part c · 4 marks

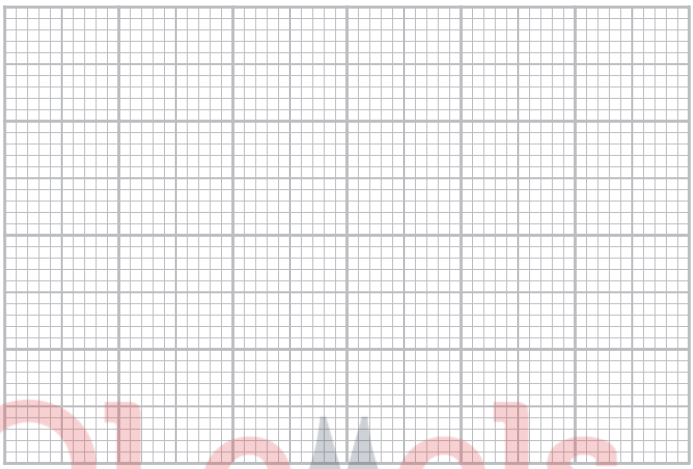
January 2022 Paper 1

Further practice · Attempt part c only.

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(i) Plot a bar chart of the results.

**(5)**



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(ii) In one year, 43 000 children are born to mothers aged 30–34.

Calculate the probable number of these children who will have a chromosome aberration.

Give your answer to two significant figures.

**(3)**

number of children = .....



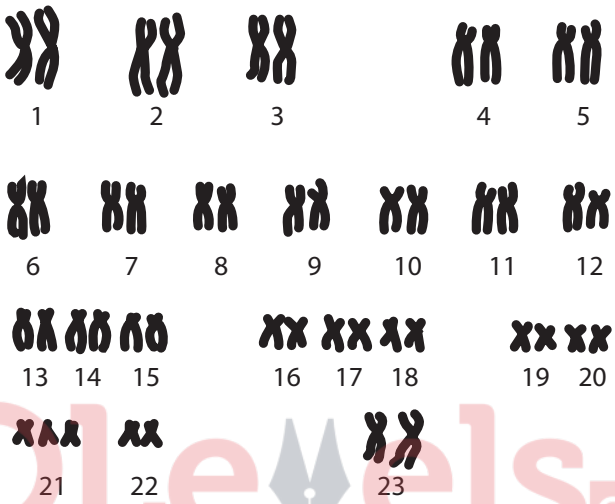
P 6 7 0 5 9 A 0 1 5 2 4

## 4. Question 5 · part c · 4 marks

January 2022 Paper 1

Further practice · Attempt part c only.

(c) The diagram shows the full diploid number of chromosomes in a person.




Justify the statement that this diagram shows the genotype of a female with a chromosome aberration.

**(4)**

**(Total for Question 5 = 15 marks)**

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## 4. Question 5 · part c · 4 marks · Working and answer

Working and answer

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# 5. Question 3 · part b.ii · 1 mark

June 2023 Paper 2R

Further practice · Attempt part b.ii only.

**3** (a) Describe what is meant by a specialised cell.

**(2)**

.....

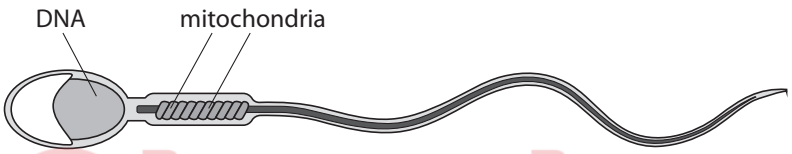
.....

.....

(b) The sperm cell is an example of a specialised cell.

The diagram shows a sperm cell.

NOT TO SCALE




(i) The sperm cell in the diagram has an actual length of 0.0060 mm.

The length of the image of the sperm cell is 84 mm.

Calculate the magnification of the sperm cell.

**(2)**

magnification = × .....

(ii) The head of the sperm contains DNA.

Give the difference in the amount of DNA found in the head of the sperm and in a nucleus of a body cell.

**(1)**

.....

.....

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P 7 2 5 8 8 A 0 8 2 4

## 5. Question 3 · part b.ii · 1 mark

June 2023 Paper 2R

Further practice · Attempt part b.ii only.

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(iii) Explain why the sperm cell has many mitochondria.

**(2)**

.....

.....

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.....

(iv) A sperm cell travels at a speed of 0.070 mm/s.

It travels a distance of 180 mm to fertilise an egg.

Use this equation to calculate the time it takes for the sperm to reach the egg.

$$\text{speed} = \frac{\text{distance}}{\text{time}}$$

Give your answer in minutes.

**(3)**

time taken = ..... minutes

(c) An electron microscope can be used to view structures in specialised cells.

Explain why an electron microscope is more useful than a light microscope to view cell structure.

**(2)**

.....

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.....

**(Total for Question 3 = 12 marks)**

## 5. Question 3 · part b.ii · 1 mark · Working and answer

Working and answer

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# 6. Question 4 · part b.iii · 3 marks

November 2023 Paper 1

Further practice · Attempt part b.iii only.

**4** The mass of DNA in a body cell changes during the cell cycle.

(a) (i) Which structure in a cell contains DNA?

**(1)**

- ☒ **A** chromosome
- ☒ **B** endoplasmic reticulum
- ☒ **C** cell membrane
- ☒ **D** ribosome

(ii) Name **one** other type of nucleic acid.

**(1)**

.....

(iii) Give **three** differences between DNA and the nucleic acid that you have named in (a)(ii).

**(3)**

| Differences | DNA | Named nucleic acid |
|---|---|---|
| 1 | | |
| 2 | | |
| 3 | | |

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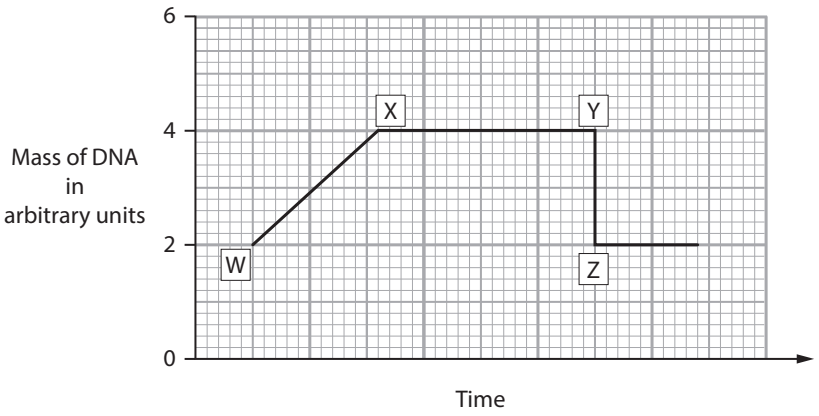
## 6. Question 4 · part b.iii · 3 marks

November 2023 Paper 1

Further practice · Attempt part b.iii only.

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(b) The graph shows the changes in the mass of DNA in a body cell, during mitosis.




(i) Describe the process that produces the change in the mass of DNA between points W and X.

**(4)**

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(ii) Describe what is happening in the cell between points Y and Z.

**(2)**

# 6. Question 4 · part b.iii · 3 marks

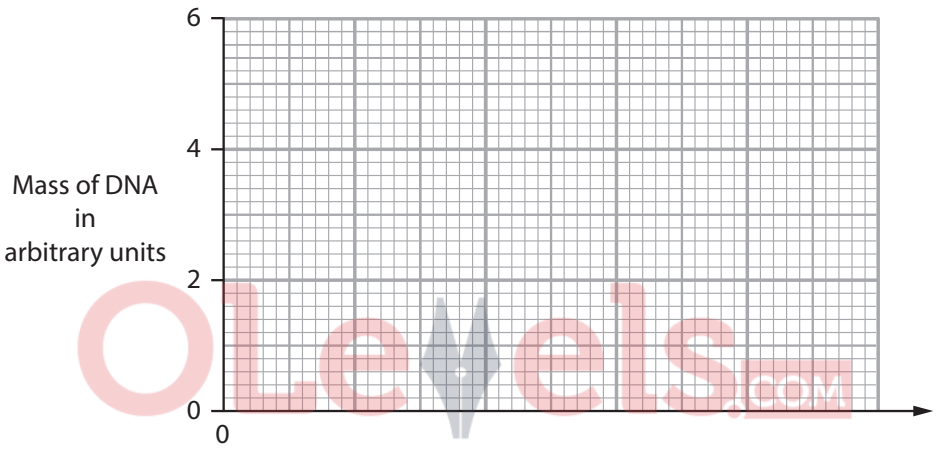
November 2023 Paper 1

Further practice · Attempt part b.iii only.

(iii) The production of gametes involves the process of meiosis.

On the grid draw a line to show the changes in the mass of DNA during meiosis.

**(3)**




**(Total for Question 4 = 14 marks)**

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Working and answer

# 7. Question 8 · part d · 3 marks

June 2019 Paper 1

Further practice · Attempt part d only.

**8** *P. vivax* is one of several parasites that can cause malaria.

(a) Explain how *P. vivax* is transmitted from one person to another.

**(3)**

(b) The table shows the total number of deaths from malaria and the number of deaths caused by *P. vivax*.

It shows the data for five different regions of the world in 2015.

| **Region of the world** | Q | R | S | T | U |
|---|---|---|---|---|---|
| **Total number of deaths from malaria** | 191 000 | 800 | 3800 | 14 400 | 1200 |
| **Number of deaths caused by *P. vivax*** | 1000 | 500 | 1400 | 4900 | 700 |

(i) Compare the number of deaths caused by *P. vivax* with the total number of deaths from malaria in regions Q and R.

Include calculations to support your answer.

**(3)**

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## 7. Question 8 · part d · 3 marks

June 2019 Paper 1

Further practice · Attempt part d only.

(ii) Suggest one reason why the number of deaths from malaria varies across the different regions.

**(1)**

(c) Scientists are developing a vaccine to protect against malaria.

Explain how vaccinating individual people will help to protect a whole population from malaria.

**(3)**

(d) *P. vivax* reproduces by asexual reproduction and sexual reproduction.

Explain why it is an advantage for a species to reproduce by both methods.

**(3)**

**(Total for Question 8 = 13 marks)**

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## 7. Question 8 · part d · 3 marks · Working and answer

Working and answer

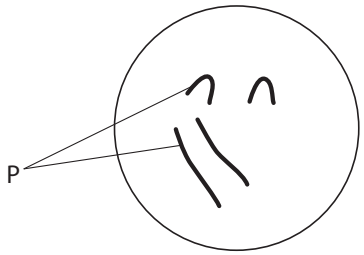
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# 8. Question 8 · part d · 4 marks

November 2020 Paper 1

Further practice · Attempt part d only.

**8** (a) The diagram shows the nucleus of a cell.



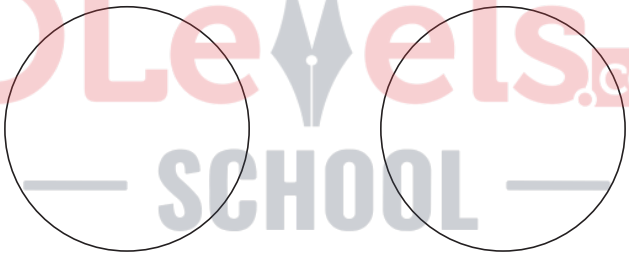

(i) Name the structures labelled P.

**(1)**

(ii) The cell divides by mitosis and two nuclei are produced.

Draw the appearance of the two nuclei.

**(2)**



(b) Before mitosis can occur, the DNA of the cell must replicate.

The table lists stages of DNA replication.

Write the numbers 1, 2, 3 and 4 in the boxes to show these stages in the correct order.

**(1)**

| Stage | Number |
|---|---|
| complementary bases pair up | |
| DNA double helix unwinds | |
| strands separate | |
| two DNA helices form | |

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# 8. Question 8 · part d · 4 marks

November 2020 Paper 1

Further practice · Attempt part d only.

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(c) The DNA of the nucleus is analysed.

The table shows the results.

| **Base** | X | adenine | Y | Z |
|---|---|---|---|---|
| **Pecentage base in DNA (%)** | 30 | 20 | 30 | 20 |

(i) Identify base Z.

**(1)**

(ii) Give the possible identities of base X.

**(1)**

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(d) Meiosis is another type of cell division.

Explain why meiosis is important in humans.

**(4)**

**(Total for Question 8 = 10 marks)**

## 8. Question 8 · part d · 4 marks · Working and answer

Working and answer

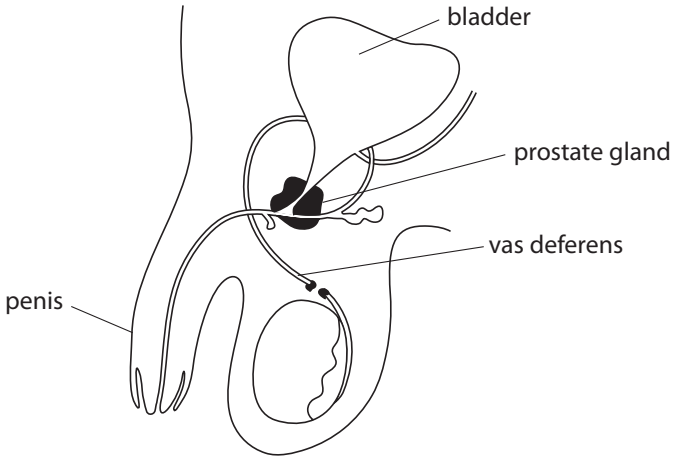
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# 9. Question 6 · part b.ii · 4 marks

January 2022 Paper 1

Further practice · Attempt part b.ii only.

**6** The diagram shows the reproductive organs of a male who has had a vasectomy.




(a) (i) Explain why having a vasectomy is an effective method of contraception.

**(4)**

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## 9. Question 6 · part b.ii · 4 marks

January 2022 Paper 1

Further practice · Attempt part b.ii only.

(ii) Explain why this method of contraception does not prevent the spread of HIV which can cause AIDS.

**(3)**

(b) (i) Add an X to the diagram to show where haploid cells are produced.

**(1)**

(ii) Explain the importance of haploid cells in the process of fertilisation.

**(4)**

**(Total for Question 6 = 12 marks)**

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## 9. Question 6 · part b.ii · 4 marks · Working and answer

Working and answer

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# 10. Question 2 · part a.iii · 3 marks

January 2023 Paper 1

Further practice · Attempt part a.iii only.

**2** (a) A human sperm cell has a tail, a mid-piece containing organelles and a head with a nucleus.

(i) Give the function of the tail.

**(1)**

(ii) Which of these, found in the mid-piece, release energy?

**(1)**

- ☒ **A** chromosomes
- ☒ **B** endoplasmic reticulum
- ☒ **C** mitochondria
- ☒ **D** ribosomes

(iii) The process of meiosis is involved in the production of sperm.

Explain the importance of meiosis in sperm production.

**(3)**

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## 10. Question 2 · part a.iii · 3 marks

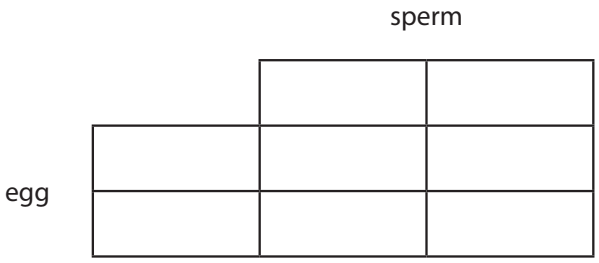
January 2023 Paper 1

Further practice · Attempt part a.iii only.

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(b) (i) Complete the genetic diagram to show how the sex of a person is determined by the chromosomes in the nucleus of the egg and the nucleus of the sperm.

**(3)**




(ii) The sperm head has to enter the egg cell so the nuclei can fuse.

Describe how the sperm head enters the egg cell.

**(2)**

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**(Total for Question 2 = 10 marks)**

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## 10. Question 2 · part a.iii · 3 marks · Working and answer

Working and answer

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# 11. Question 4 · part b.ii · 4 marks

June 2024 Paper 2R

Further practice · Attempt part b.ii only.

**4** (a) A student does a survey of eye colour in a sample of people.

The table shows a tally of the number of people with each eye colour in the sample.

| Blue | Green | Brown | Hazel | Grey |
|---|---|---|---|---|
| 𝍸 𝍸 𝍸 𝍸 𝍸 | 𝍸 𝍸 I | 𝍸 𝍸 𝍸 𝍸 𝍸 I | 𝍸 𝍸 𝍸 𝍸 𝍸 | III |

(i) Complete the table by giving the missing information.

**(1)**

| Number of people with each eye colour | | | | |
|---|---|---|---|---|
| **Blue** | **Green** | **Brown** | **Hazel** | **Grey** |
| 25 | | | | 3 |

(ii) Calculate the percentage of people with brown eyes in the sample.

**(2)**

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percentage = ..... %

(b) Eye colour is determined by alleles inherited from parents.

(i) Which of these gives the definition of an allele?

**(1)**

- ☒ **A** two dominant genes in a genotype
- ☒ **B** two recessive genes in a genotype
- ☒ **C** different genes in a genotype
- ☒ **D** alternative forms of the same gene in a genotype

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## 11. Question 4 · part b.ii · 4 marks

June 2024 Paper 2R

Further practice · Attempt part b.ii only.

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(ii) Explain why diploid cells contain two alleles for each characteristic.

**(4)**

(c) Cystic fibrosis is a condition that is also determined by alleles that are inherited from parents.

(i) Give the possible genotypes of two parents that have a child with cystic fibrosis.

**(2)**

(ii) Give one symptom of cystic fibrosis.

**(1)**

**(Total for Question 4 = 11 marks)**

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## 11. Question 4 · part b.ii · 4 marks · Working and answer

Working and answer

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Practice workbook · Name contraceptive methods, explain how they work and compare them

Barrier, hormonal, natural, IUD and surgical methods: naming examples, explaining the mechanism, limits such as infection protection, and weighing two methods with data.

8 question parts: 1 worked-example retry and 7 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-CONTRACEPTION-METHODS-AND-ACTION

# 1. Question 1 · part b.i · 3 marks

November 2024 Paper 2

Worked-example retry · Attempt part b.i only.

**1** (a) The information in the box gives structures or hormones involved in reproduction.

| | | | | |
|---|---|---|---|---|
| cervix | FSH | ovary | oviduct | ovum |
| oxytocin | progesterone | uterus | vagina | vulva |

The table lists some of the functions of these structures and hormones.

Complete the table by giving the missing information.

**(6)**

| Function | Name of structure or hormone |
|---|---|
| transfers ova to uterus | |
| contracts during birth | |
| releases ova | |
| dilates to allow birth | |
| maintains uterus lining during pregnancy | |
| causes contractions during birth | |

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# 1. Question 1 · part b.i · 3 marks

November 2024 Paper 2

Worked-example retry · Attempt part b.i only.

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(b) There are various methods of birth control.

(i) Complete the table by giving an example of each method.

**(3)**

| Method of birth control | Example |
|---|---|
| barrier | |
| natural | |
| hormonal | |

(ii) Name one other method of birth control not given in the table.

**(1)**

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**(Total for Question 1 = 10 marks)**

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Working and answer

## 2. Question 4 · part a · 3 marks

June 2022 Paper 2R

Further practice · Attempt part a only.

**4** (a) State three methods of birth control.

**(3)**

1 .....

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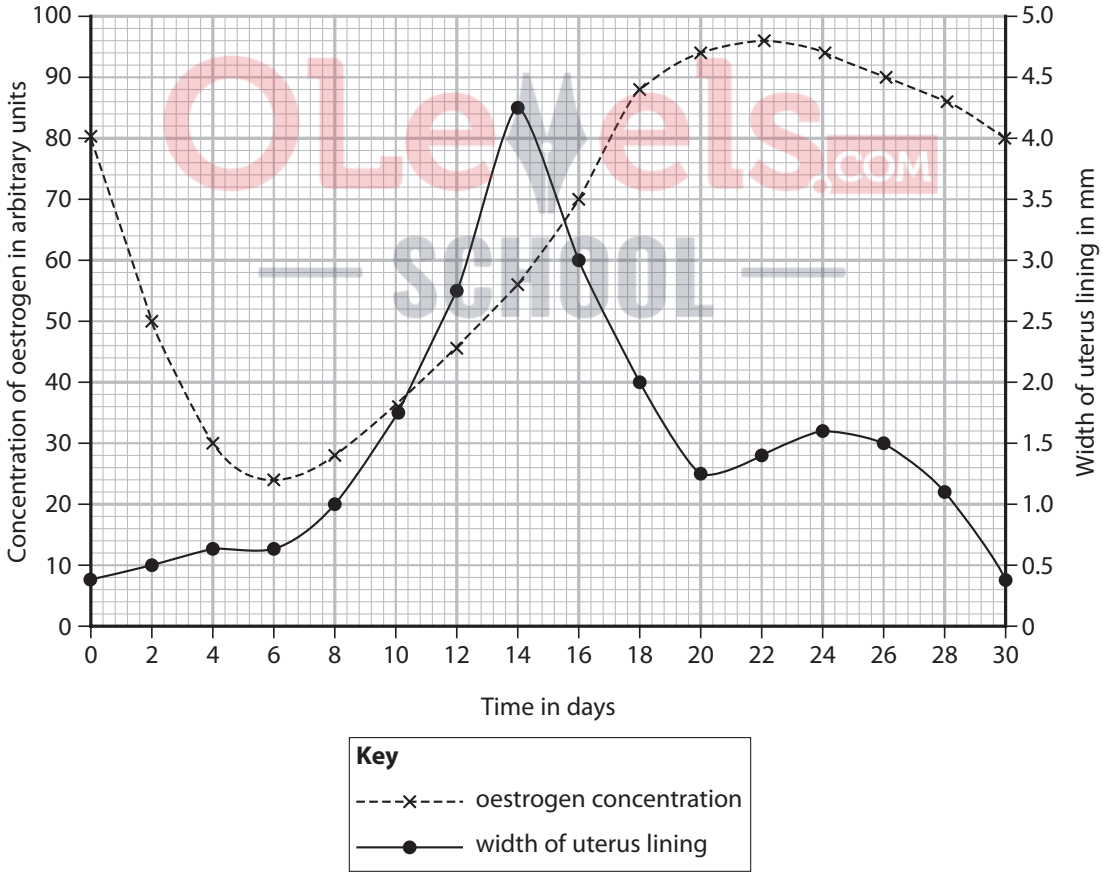
2 .....

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3 .....

.....

(b) The graph shows the changes in the width of the uterus lining and in the concentration of oestrogen in the blood during one menstrual cycle.




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## 2. Question 4 · part a · 3 marks · Working and answer

Working and answer

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## 3. Question 6 · part b · 1 mark

June 2024 Paper 1R

Further practice · Attempt part b only.

**6** (a) The table gives the names and functions of hormones in the menstrual cycle.

Complete the table by placing one tick (✓) in each row of the table to show the main function of each hormone in the menstrual cycle.

**(3)**

| Hormone | Function of hormone | | | |
|---|---|---|---|---|
| | causes ovulation | builds up uterus lining after menstruation | maintains uterus lining during pregnancy | stimulates a follicle to mature |
| progesterone | | | | |
| FSH | | | | |
| oestrogen | | | | |

(b) The chance of pregnancy can be decreased by using contraception.

State **one** method of barrier contraception.

**(1)**

.....

(c) In-vitro fertilisation (IVF) can be used to increase the chances of pregnancy.

Describe the process of IVF.

**(4)**

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## 3. Question 6 · part b · 1 mark · Working and answer

Working and answer

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# 4. Question 1 · part b.ii · 1 mark

November 2024 Paper 2

Further practice · Attempt part b.ii only.

**1** (a) The information in the box gives structures or hormones involved in reproduction.

| | | | | |
|---|---|---|---|---|
| cervix | FSH | ovary | oviduct | ovum |
| oxytocin | progesterone | uterus | vagina | vulva |

The table lists some of the functions of these structures and hormones.

Complete the table by giving the missing information.

**(6)**

| Function | Name of structure or hormone |
|---|---|
| transfers ova to uterus | |
| contracts during birth | |
| releases ova | |
| dilates to allow birth | |
| maintains uterus lining during pregnancy | |
| causes contractions during birth | |

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## 4. Question 1 · part b.ii · 1 mark

November 2024 Paper 2

Further practice · Attempt part b.ii only.

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(b) There are various methods of birth control.

(i) Complete the table by giving an example of each method.

**(3)**

| Method of birth control | Example |
|---|---|
| barrier | |
| natural | |
| hormonal | |

(ii) Name one other method of birth control not given in the table.

**(1)**

.....

**(Total for Question 1 = 10 marks)**

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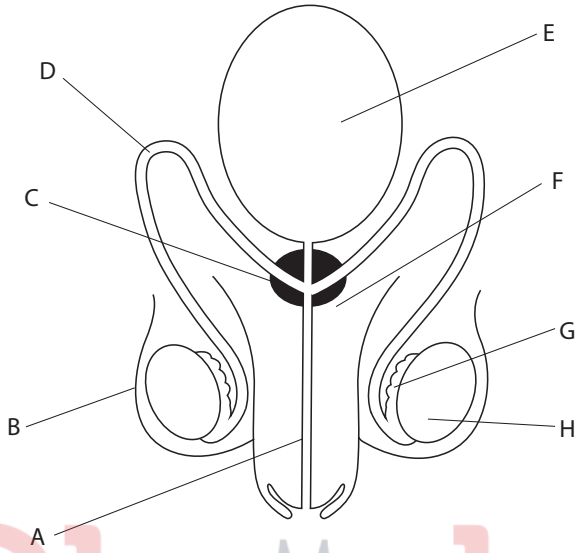
Working and answer

# 5. Question 7 · part c.ii · 1 mark

June 2019 Paper 2R

Further practice · Attempt part c.ii only.

**7** (a) The diagram shows the male reproductive system.




The table gives descriptions of structures in the male reproductive system.

Complete the table by giving a letter from the diagram to match each description.

Each letter may be used once, more than once or not at all.

**(4)**

| Description of the structure | Letter |
|---|---|
| stores sperm | |
| produces sperm | |
| produces fluid to help sperm move | |
| contains cells dividing by meiosis | |

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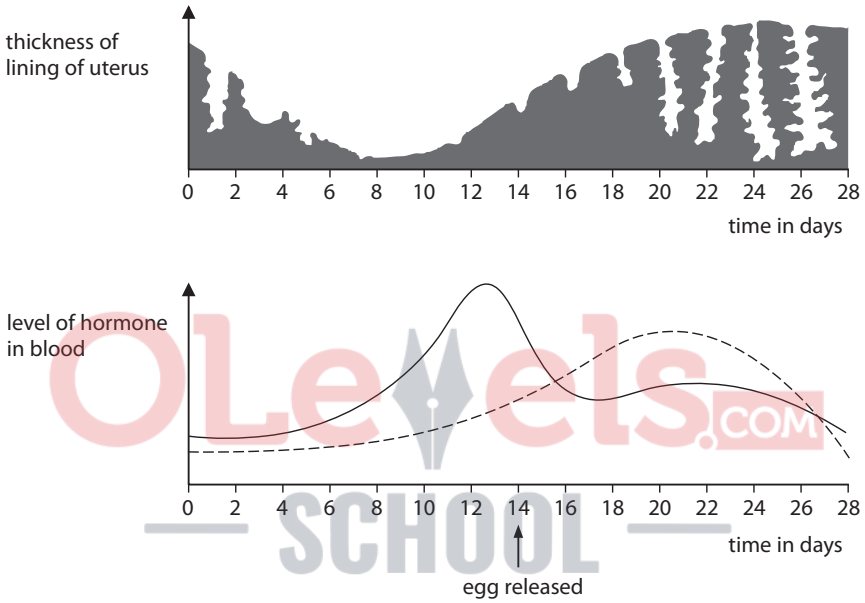
# 5. Question 7 · part c.ii · 1 mark

June 2019 Paper 2R

Further practice · Attempt part c.ii only.

(b) Changes occur in the female body during the menstrual cycle.

The graphs show changes in the lining of the uterus and the levels of two hormones that control these changes.




(i) Use information from the graphs to explain the cause of growth in the lining of the uterus after day 12.

**(2)**

(ii) Explain why fertilisation is less likely to occur from intercourse between days 8 and 12.

**(2)**

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# 5. Question 7 · part c.ii · 1 mark

June 2019 Paper 2R

Further practice · Attempt part c.ii only.

(c) A synthetic oestrogen called oestradiol is used to make contraceptive pills. Natural oestrogen, which is a lipid molecule, could be used but is less effective than oestradiol when taken as a pill.

(i) Suggest why natural oestrogen is not as effective as oestradiol when taken as a pill.
**(2)**

(ii) Give a reason why contraceptive pills only need to be taken for 21 days out of the 28-day cycle.
**(1)**

**(Total for Question 7 = 11 marks)**

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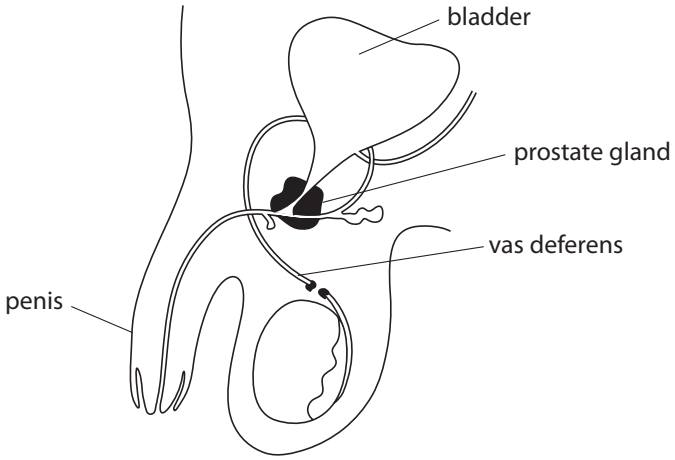
Working and answer

# 6. Question 6 · part a.i · 4 marks

January 2022 Paper 1

Further practice · Attempt part a.i only.

**6** The diagram shows the reproductive organs of a male who has had a vasectomy.




(a) (i) Explain why having a vasectomy is an effective method of contraception.

**(4)**

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# 6. Question 6 · part a.i · 4 marks

January 2022 Paper 1

Further practice · Attempt part a.i only.

(ii) Explain why this method of contraception does not prevent the spread of HIV which can cause AIDS.

**(3)**

(b) (i) Add an X to the diagram to show where haploid cells are produced.

**(1)**

(ii) Explain the importance of haploid cells in the process of fertilisation.

**(4)**

**(Total for Question 6 = 12 marks)**

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## 6. Question 6 · part a.i · 4 marks · Working and answer

Working and answer

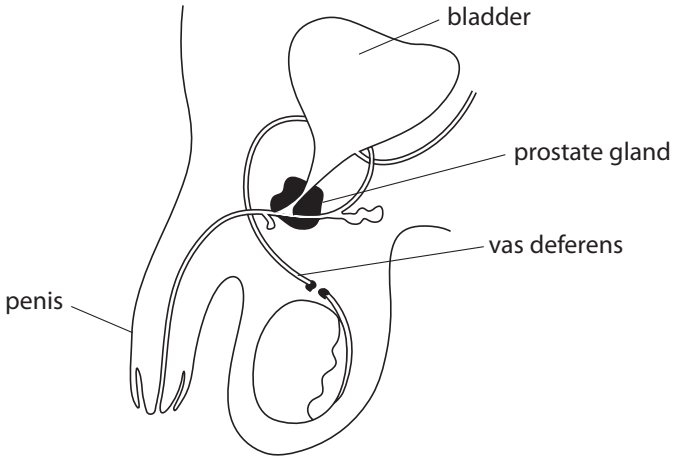
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# 7. Question 6 · part a.ii · 3 marks

January 2022 Paper 1

Further practice · Attempt part a.ii only.

**6** The diagram shows the reproductive organs of a male who has had a vasectomy.




(a) (i) Explain why having a vasectomy is an effective method of contraception.

**(4)**

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## 7. Question 6 · part a.ii · 3 marks

January 2022 Paper 1

Further practice · Attempt part a.ii only.

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(ii) Explain why this method of contraception does not prevent the spread of HIV which can cause AIDS.

**(3)**

(b) (i) Add an X to the diagram to show where haploid cells are produced.

**(1)**

(ii) Explain the importance of haploid cells in the process of fertilisation.

**(4)**

**(Total for Question 6 = 12 marks)**

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## 7. Question 6 · part a.ii · 3 marks · Working and answer

Working and answer

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## 8. Question 5 · part whole · 5 marks

June 2022 Paper 1R

Further practice · Attempt part whole only.

**5** The table gives information about two methods of contraception.

| Method | Percentage effectiveness (%) | Information |
|---|---|---|
| male condom | 82 | • worn during sexual intercourse<br>• may slip off or tear during sexual intercourse |
| sterilisation | 99 | • surgery required<br>• permanent method |

Evaluate these two methods of contraception using information from the table and your own knowledge.

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**(Total for Question 5 = 5 marks)**

## 8. Question 5 · part whole · 5 marks · Working and answer

Working and answer

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